High-dimensional state trajectories reveal transformer inference dynamics

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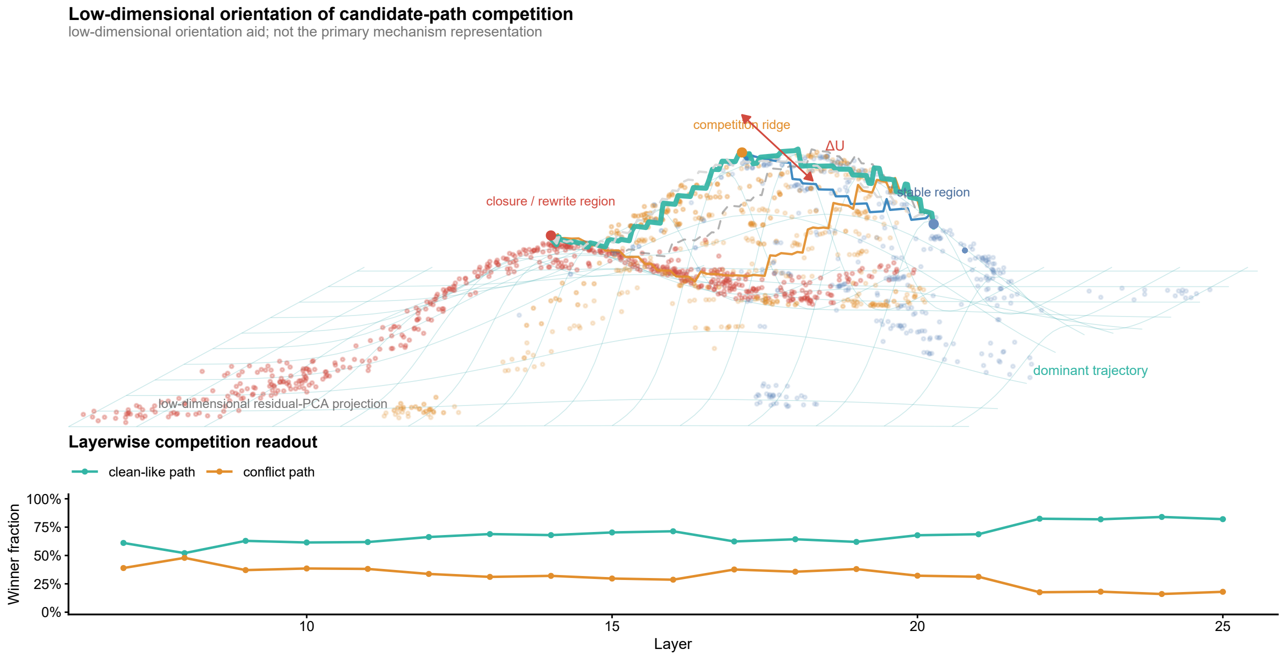
Mechanistic studies localize language-model function in parameters, heads or neurons, but these parts do not show how one inference unfolds. We treat the ordered layer-state trajectory as the empirical process. Across Qwen, Llama and Gemma, realized order was 1.60-2.24 times more chord-aligned than endpoint-preserving shuffles. Random initializations were still more aligned, whereas reordered executions retained geometric regularity but lost native outputs: architecture supplies a propagation scaffold, but order is functionally consequential. In a Pythia-160m training audit, fixed task coordinates emerged; task function required compatible input, block and output components. Ordered-prefix summaries did not generally predict final boundaries beyond current state. After additive interventions failed, confidence-gated local-transition actuators crossed a declared full-vocabulary boundary in 17/87, 13/87 and 9/87 held-out prompts, with 0/203, 2/203 and 0/203 non-target changes. Ordered high-dimensional states make inference observable and locally controllable within a relation-graph task family; geometry, forecasting, action assignment and correctness remain separate limits.



Auxiliary visual orientation. Three-dimensional residual-PCA projection of candidate-path competition. This low-dimensional rendering is included only to aid intuition; it is not a literal state-space manifold or independent mechanism evidence.

Large language models produce ordered linguistic and reasoning behaviour despite computations distributed across many parameters and layers[1-4]. Mechanistic interpretability has made those computations experimentally accessible, yet a common description of how an inference unfolds remains elusive[13,14]. The usual explanation is scale. We considered a prior possibility: the difficulty may begin before measurement, in the choice of what is being measured. A mechanism can be implemented by parameters, attention heads, multilayer perceptrons and residual blocks without being most simply expressed as any one of those engineering objects.

Probes establish what information is available in activations, provided that the probe itself is controlled[5,6]. Head and circuit analyses identify local computational motifs[7-9,18]; causal tracing and editing localize consequential associations[10]; vocabulary readbacks expose latent output tendencies[11,19,20]; and sparse-feature methods seek more interpretable activation units[12,21,22]. Each approach answers an important question about an implementing component or an observable. Their natural engineering boundaries, however, do not necessarily define the empirical object on which an inference mechanism is comparable across depth, checkpoints and interventions.

This distinction suggests a methodological ordering: choose a suitable empirical research object before choosing its coordinates. We therefore switched from a catalogue of implementing parts to the ordered inference process and represented it by the sequence of high-dimensional layer states. Layer depth is an ordered evolution coordinate, not physical time.

Comparable methodological choices arise in other high-dimensional systems. Physical dynamics may require informative state or reaction coordinates, whereas systems neuroscience can reveal population trajectories that are not apparent from single-unit descriptions[23-25]. We use these as sources of testable questions, not as evidence that transformers share physical or neural mechanisms.

Existing representation-comparison methods ask related but different questions. SVCCA and related similarity measures compare representation sets through chosen invariances and geometries[15,16]; representation-manifold paths compare how networks transform data across depth[17]. Our unit is one prompt-conditioned inference trajectory. We retain its realized block order, perturb that order under a fixed-endpoint null and intervene on its local transitions. Vocabulary projections and compressed views can therefore be audited as coordinates of this process rather than assumed to be the process itself.

We asked whether this object makes otherwise separate observations commensurable. We audited direct states and flows against vocabulary readbacks; used exploratory model-specific windows to construct bounded candidate-path separation; separated architectural continuity from training-organized function; and used intervention, including failed actuators, as a perturbational test. The resulting chain establishes an observable process and bounded control of a declared emitted-token boundary within one relation-graph task family across all three checkpoints. It does not establish a complete predictive theory, exact geodesic law or answer-correctness control.

Layer-state trajectories expose an ordered process

For checkpoint m, prompt p and layer l, let $h_{m,p,l} \in \mathbb{R}^{d_m}$ be the last-token hidden state and

$$T_{m,p} = (h_{m,p,1}, \ldots, h_{m,p,L_m})$$

the ordered inference trajectory. The definition preserves two features that disappear when a model is reduced to a parts list: the state visited at each depth and the order in which those states are visited. We used parameters and modules to extract the trajectory, but treated the trajectory itself as the empirical process object (Fig. 1a). Its high dimensionality was retained in the analyses; the low-dimensional orientation view in Extended Data Fig. 1 is not evidentiary.

The object switch changes the question from which component contains a feature to how a prompt-conditioned state evolves through the components that implement it. Adjacent hidden-state cosine distances changed non-uniformly with depth, and the normalized participation ratio of each update showed that its energy remained distributed across many hidden dimensions (Fig. 1b,c). These measurements use native normalized states rather than a vocabulary projection. They establish an ordered, distributed process object, not continuous physical time or checkpoint-identical coordinates.

Each prompt presented a relation graph and required one of two one-word labels. "Ava belongs to Bela" and "Bela maps to Red" imply Red for Ava. Red is the clean candidate from the unperturbed chain; an added statement introduces Blue as the conflict candidate. Before fitting, prompt metadata were collapsed into three analysis labels: stable prompts leave the original route decisive, competition prompts support both labels, and closure prompts update or override the route, allowing the conflict candidate to become rule-consistent. These are predefined prompt-condition classes, not dynamical states discovered from the trajectories. Clean and conflict mark provenance, not correctness. Neither ΔU nor model outputs entered these labels; the coordinate-audit and intervention panels used separately fixed mappings (Supplementary Methods Table 1), and the testbed does not represent open-ended reasoning.

We next audited the coordinates used to observe that process. In a five-fold relation-graph-grouped test with coordinate-audit class labels fixed by prompt condition, hidden decision state, hidden precursor state and direct hidden flow increased macro F1 by 0.448, 0.448 and 0.444 in Qwen; 0.520, 0.520 and 0.485 in Llama; and 0.240, 0.248 and 0.244 in Gemma. Learned-output-matrix readbacks increased it by 0.407, 0.268 and 0.217, respectively (Fig. 2a). The advantage was checkpoint-dependent but did not require ΔU. Readbacks remain informative output-facing diagnostics, but their geometry depends on the projection matrix and selection rule. Real, row-permuted and Gaussian projection controls did not give the learned readback the preservation advantage expected of a primary process coordinate (Fig. 2b).

The decision-window clean-relative margin profile is retained separately as a construction audit (Extended Data Fig. 2). Because ΔU is the first principal component of the same matrix, that profile is not independent evidence. Removing this circular comparison leaves the group-held-out mechanism-label result and projection controls as the basis of the coordinate audit.

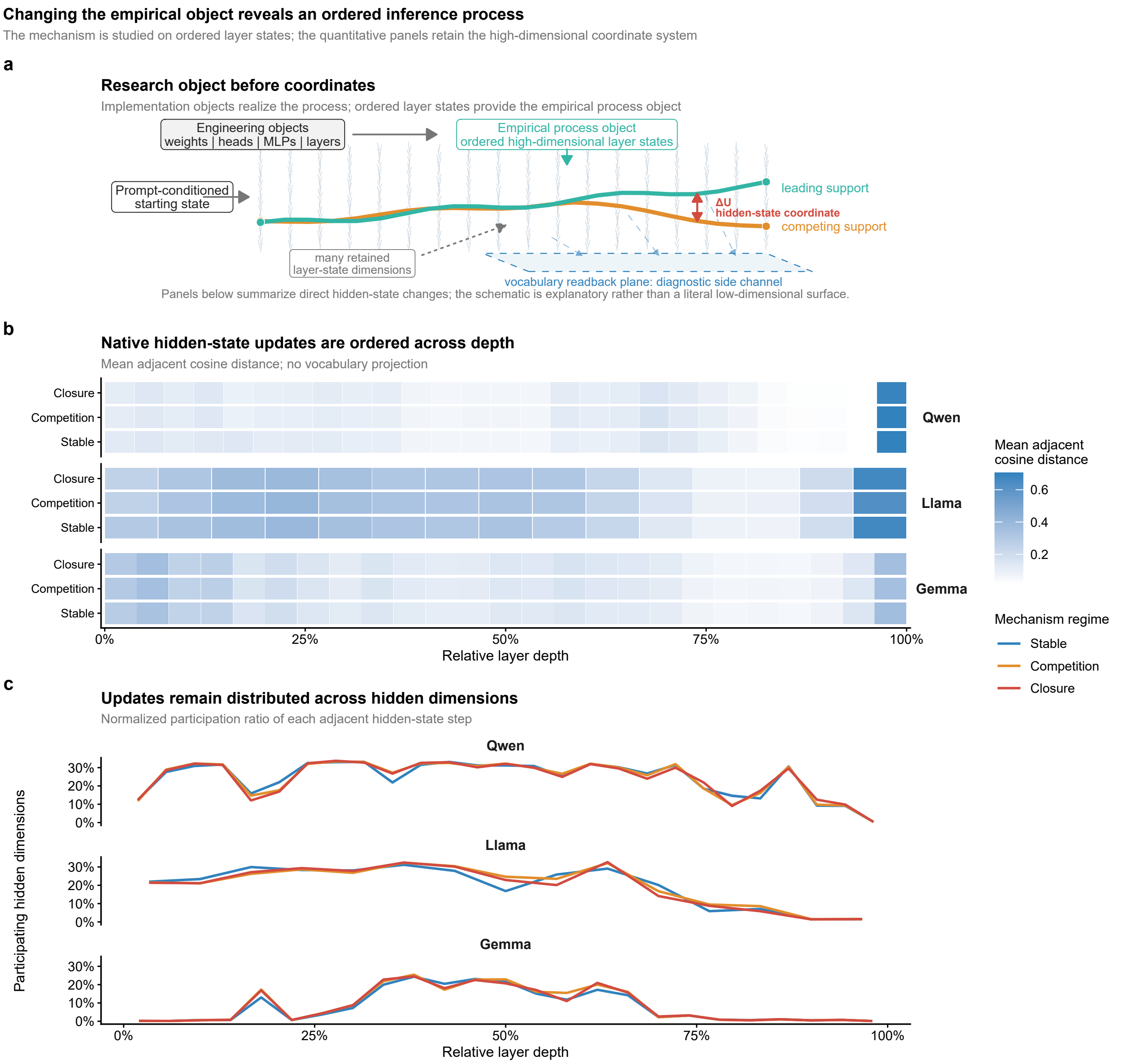


Fig. 1 | Changing the empirical object reveals an ordered inference process. a, Distinction between implementation objects, including weights, attention heads, MLPs and layers, and the empirical process object used here: the ordered high-dimensional layer-state trajectory. b, Mean adjacent cosine distance between normalized hidden states across relative depth and collapsed prompt-condition families. c, Normalized participation ratio of adjacent hidden-state steps, reporting the fraction of hidden dimensions over which step energy is distributed. Panels b and c use direct hidden states without vocabulary projection. The condition families are predefined analysis labels, not trajectory-derived phases. Layer depth is an ordered evolution coordinate, not physical time. The schematic does not depict a literal low-dimensional surface. See Source Data Fig. 1 and Source Data manifest; model and extraction metadata are in Extended Data Table 1.

Exploratory registration bounds candidate-path competition

Absolute layer number is a poor default correspondence across checkpoints with different depths and internal schedules. We therefore used functional registration as an exploratory alternative to layer-number matching. Sliding windows within the first 45% of depth were scored for clean-relative preservation, prompt-condition-class information and coupling to a later dominance coordinate. The selected intervals differed across checkpoints: 15-35% relative depth for Qwen, 0-25% for Llama and layers 0-6 for Gemma (Fig. 2c,d). Their scan scores supplied model-specific analysis windows; they did not establish homologous locations or shared coordinates.

These windows are exploratory and data-selected. The same scan supplied both candidate windows and the score used to select them. Grouped cross-validation protected the within-window classification and correlation estimates from group overlap, but selection was not nested. We subsequently froze the selected windows and tested 24 new graph groups with disjoint entities, relation phrases and candidate labels. Prompt-condition-class information transferred in all three checkpoints (macro F1 0.739, 0.662 and 0.649; each above its label-permutation null), whereas downstream coupling exceeded its target-permutation null in Qwen and Gemma but not Llama. A predeclared topology-separation test reversed sign in all three checkpoints, and only Gemma retained a top-quartile frozen-window rank (Supplementary Methods Table 2). The three-checkpoint confirmation gate therefore failed. Together with weak pairwise direction isomorphism (Extended Data Fig. 2), this leaves the windows as model-specific exploratory indexing choices, not confirmed functional phases.

Within the controlled tasks, the clean-relative signed margin $dR_{p,l}$ compared support for the task-defined clean and conflict candidates relative to the clean condition in the same graph group. Principal-component analysis reduced the decision-window matrix $D = [dR_{p,l}]$, and its oriented first-component score defined ΔU (Fig. 3a). This task-constructed exploratory diagnostic describes separation between declared candidate supports; it is not an independently discovered latent variable. It can be computed from a trajectory, audited under information-removal controls and used as a held-out intervention endpoint. Reaction-coordinate compression motivated the construction, but ΔU is not an energy, physical potential or universal order parameter.

The correlations between ΔU and the related dominance readout (0.9990, 0.9989 and 0.9997) are construction checks, not independent evidence. Predecision recomputation excluded decision-layer and endpoint features; candidate-margin-excluded flow removed the direct margin; topology-window and auxiliary-band controls changed the interval (Fig. 3b-d). Mechanism macro F1 ranged from 0.771 to 0.954, and PC1 explained 0.832-0.967 of decision-profile variance. These tests ask whether accompanying trajectory signatures survive removal or displacement of construction information; they do not establish out-of-task generalization. Final correctness did not enter the construction. We therefore treat ΔU as bounded in inferential role, not numerical range.

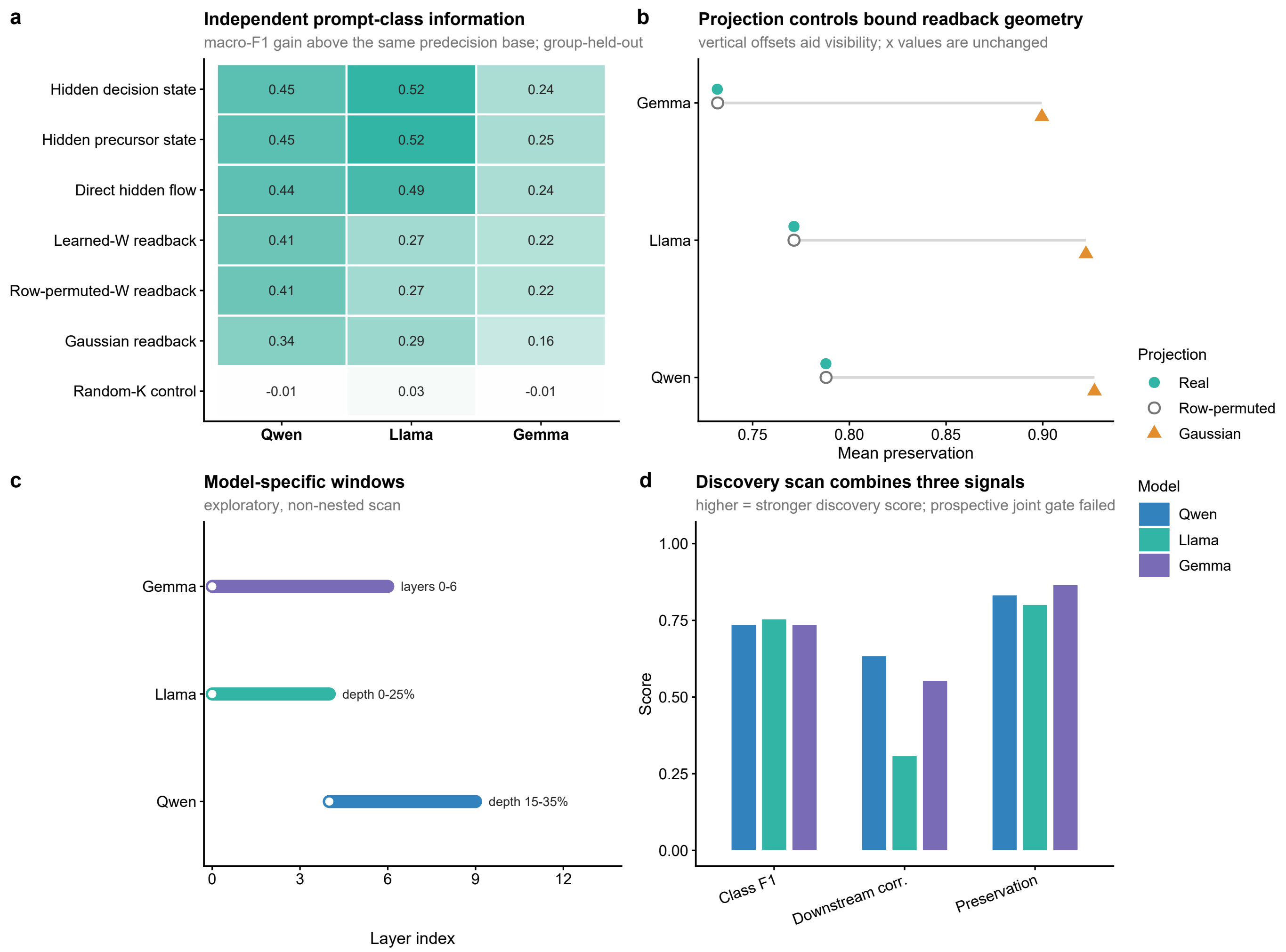


Fig. 2 | Coordinate audit and exploratory functional registration. a, Group-held-out prompt-condition-class macro-F1 increment above the same predecision base; positive values indicate additional information. Reduction was fitted within each training fold. b, Mean readback-geometry preservation under real, row-permuted and Gaussian projections; vertical offsets only separate overlapping points. c, Discovery-scan windows as layer or relative-depth intervals. d, Prompt-class F1, downstream correlation and clean-relative preservation used in the scan; higher values are stronger discovery scores. The construction-aligned profile is excluded from a because it shares its matrix with ΔU (Extended Data Fig. 2). Readbacks remain diagnostics. A prospective 24-graph test failed the joint three-checkpoint recurrence gate, so the windows remain exploratory (Supplementary Methods Table 2). See Source Data Fig. 2 and Source Data manifest.

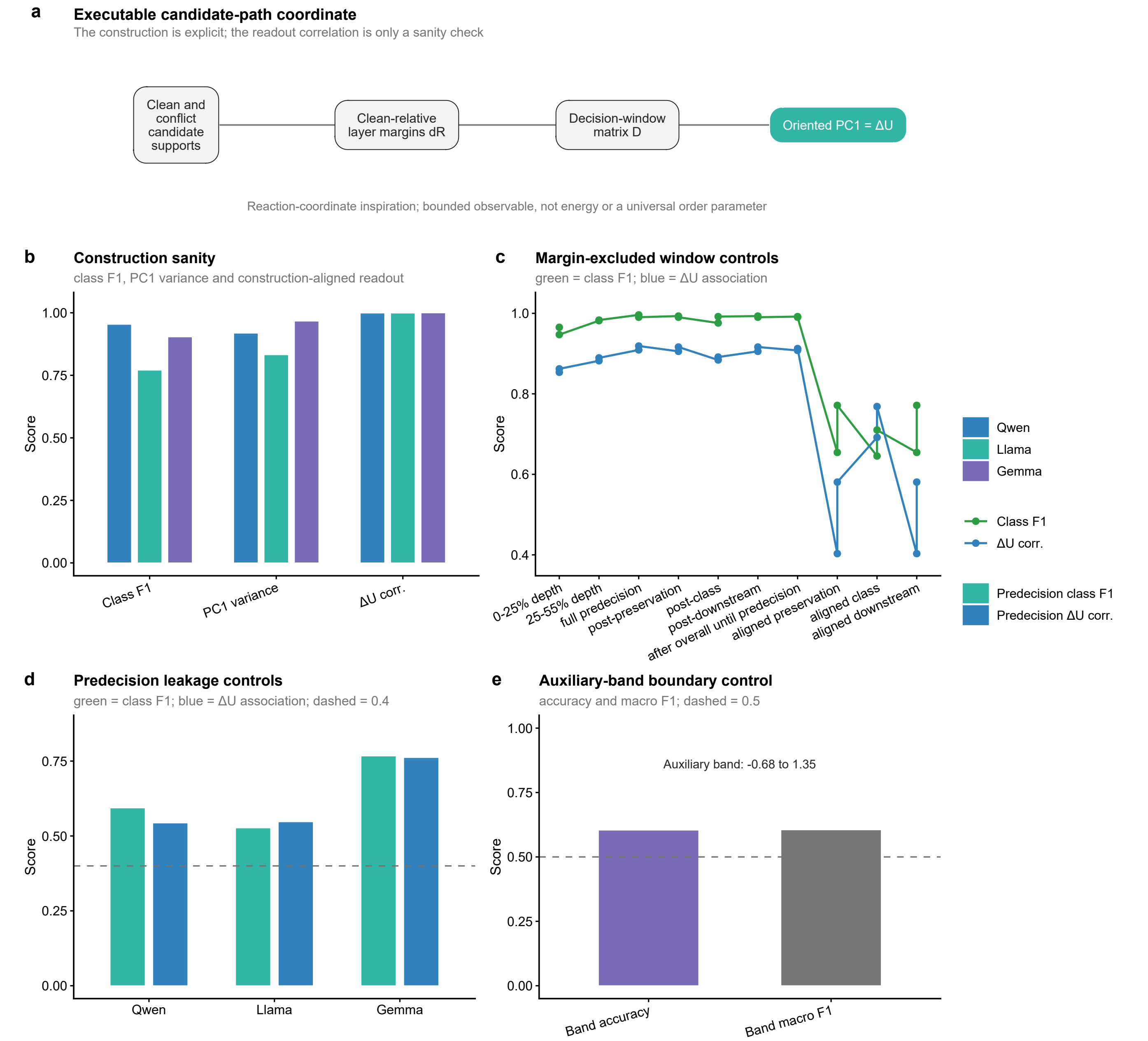


Fig. 3 | Construction and bounds of candidate-path separation. a, Candidate supports define clean-relative layer margins dR, the decision-window matrix D and oriented PC1 score ΔU. b, Prompt-class F1, PC1 variance and the construction-aligned ΔU-dominance check. c,d, Candidate-margin-excluded and predecision controls; green is class F1, blue is ΔU association, and higher values indicate greater retention of the original diagnostic. e, Auxiliary-band accuracy and macro F1. Dashed lines mark 0.4 in d and 0.5 in e. ΔU is a bounded candidate-path separation observable, not an energy or universal order parameter. See Source Data Fig. 3 and Source Data manifest.

Layer order separates geometric scaffold from function

A useful process object should do more than separate conditions. It should also make observed updates compactly describable. We compared state-only features, previous-transition history features, state plus a realized continuous transition coordinate O_l^{cont}, and a bilinear state-transition model (Fig. 4a). Previous-transition history features summarize reversals, changes in relative rank and backtracking before the current transition. They are distinct from the decision-window clean-relative margin profile.

The realized coordinate O_l^{cont} comprises continuous components of the observed adjacent transition. Adding it raised macro R^2 to 0.980, and the bilinear state-transition variant reached 0.993. This is descriptive closure because the realized coordinate contains target-transition information. In a separate pre-transition test, the state-only model reached 0.822, the observed-coordinate model reached 0.979 and the best estimated-coordinate bilinear model reached 0.862, retaining 25.1% of the observed-coordinate gain (Extended Data Fig. 4). Across 768 held-out prompt groups, its macro-R^2 increment over state only was 0.0394 (group-bootstrap 95% interval, 0.0368-0.0426). History and context predicted the coordinate itself with macro R^2 values of 0.41-0.61. Thus target-excluded estimates carry finite prospective information, but O_l^{cont} still compactly factorizes the realized update far better than the estimated coordinate and does not yet supply an equivalent forward law.

We therefore tested a narrower prospective question without using O_l^{cont}: can an ordered partial trajectory forecast the exact final clean-minus-conflict output margin for unseen graph groups? Ridge models used only the current and earlier decision-window margins, with 67 graph groups for fitting and 29 held out. At the earliest cutoff whose held-out R^2 exceeded the 95th percentile of 50 independent within-prompt order nulls, R^2 was 0.574, 0.393 and 0.714 for Qwen, Llama and Gemma; boundary AUROC was 0.857, 0.837 and 0.984 (Extended Data Fig. 6c). This is a bounded forecast of one declared candidate margin. A stronger cross-task test asked whether ordered-prefix summaries added held-out information beyond the current state: 6 of 18 initial conditions passed, all in addition; lexical prompts passed 0/9 and an independent mixed-arithmetic replication passed 0/9 (Extended Data Fig. 7c). Thus the tested histories do not establish a general predictive law; the current high-dimensional state may already summarize much of the usable path history.

We next tested finite-path directional organization in the native hidden-state trajectories. States were normalized to unit length; adjacent steps were compared with the endpoint chord, and detour and turning curvature were calculated. Each of 50 checkpoint-level null repeats preserved both endpoints and all intermediate states but applied one shared random layer permutation to the 96-prompt controlled subset. Observed order was 1.60-2.24 times more chord-aligned than the null mean across Qwen, Llama and Gemma, corresponding to 5.11-5.63 null standard deviations; detour and turning curvature were also reduced (Fig. 4b,c).

This order effect recurred in controlled lexical and addition tasks and a 96-item four-choice ARC-Challenge audit without binary reduction[26]: real order exceeded all 50 shuffles in every task-checkpoint combination (null-standardized effects, 3.21-7.84; Extended Data Fig. 6a,b). Yet three random models from each checkpoint configuration showed still larger alignment gains, and stronger relational-continuity nulls passed in all 18 architecture-task-seed conditions (Extended Data Fig. 7a). Chord-directed order therefore exposes an architecture-compatible propagation scaffold, not learned inference by itself.

Geometry was not sufficient for function. In Qwen, reversed and fixed-permutation executions retained positive chord contrasts but had zero full-vocabulary top-1 agreement with native execution, candidate agreement of 0.479 and mean Jensen-Shannon divergence of 0.692 (Extended Data Fig. 7b). In Llama, direction-only trajectories passed 0/7 task gates whereas norm-only fixed-direction trajectories passed 7/7, and radial rescaling changed the metric monotonically in all seven tests (Extended Data Fig. 7d). Native order is functionally consequential, but raw alignment is scale-sensitive and cannot stand in for learned computation.

To isolate what training adds, we swapped embeddings (E), blocks (B), final normalization (N) and output head (W) between initialized and trained Pythia-160m checkpoints. On 53 held-out lexical cloze prompts, accuracy rose from 0.472 to 0.849 when all components were trained; no isolated trained component exceeded 0.604, and the superadditive margin excess was 2.621 (95% confidence interval, 1.252-3.987; Extended Data Fig. 8a,b). Across seven checkpoints, model-native candidate-coordinate order gain emerged from step 1,000 and correlated with accuracy (Spearman $\rho = 0.75$). A coordinate fitted at the final checkpoint transferred poorly at steps 0 and 128, reached macro F1 0.701 at step 1,000 and 1.0 by step 16,000. The within-checkpoint category control was already perfect at initialization and was rejected as evidence of learning (Extended Data Fig. 8c,d). Training therefore organizes compatible input, transition and output coordinates rather than placing task function in any isolated component.

These metrics describe directional efficiency under a specified cosine geometry and shuffle null; they neither learn a Riemannian metric nor test a geodesic equation. Residual-only, parallel and perpendicular proxies remained complementary (Fig. 4d,e), excluding a one-dimensional collapse. We therefore use chord-directed as the empirical term and withdraw geodesic-biased as a result label; vocabulary-readback-centre trajectories remain auxiliary diagnostics (Extended Data Fig. 3c).

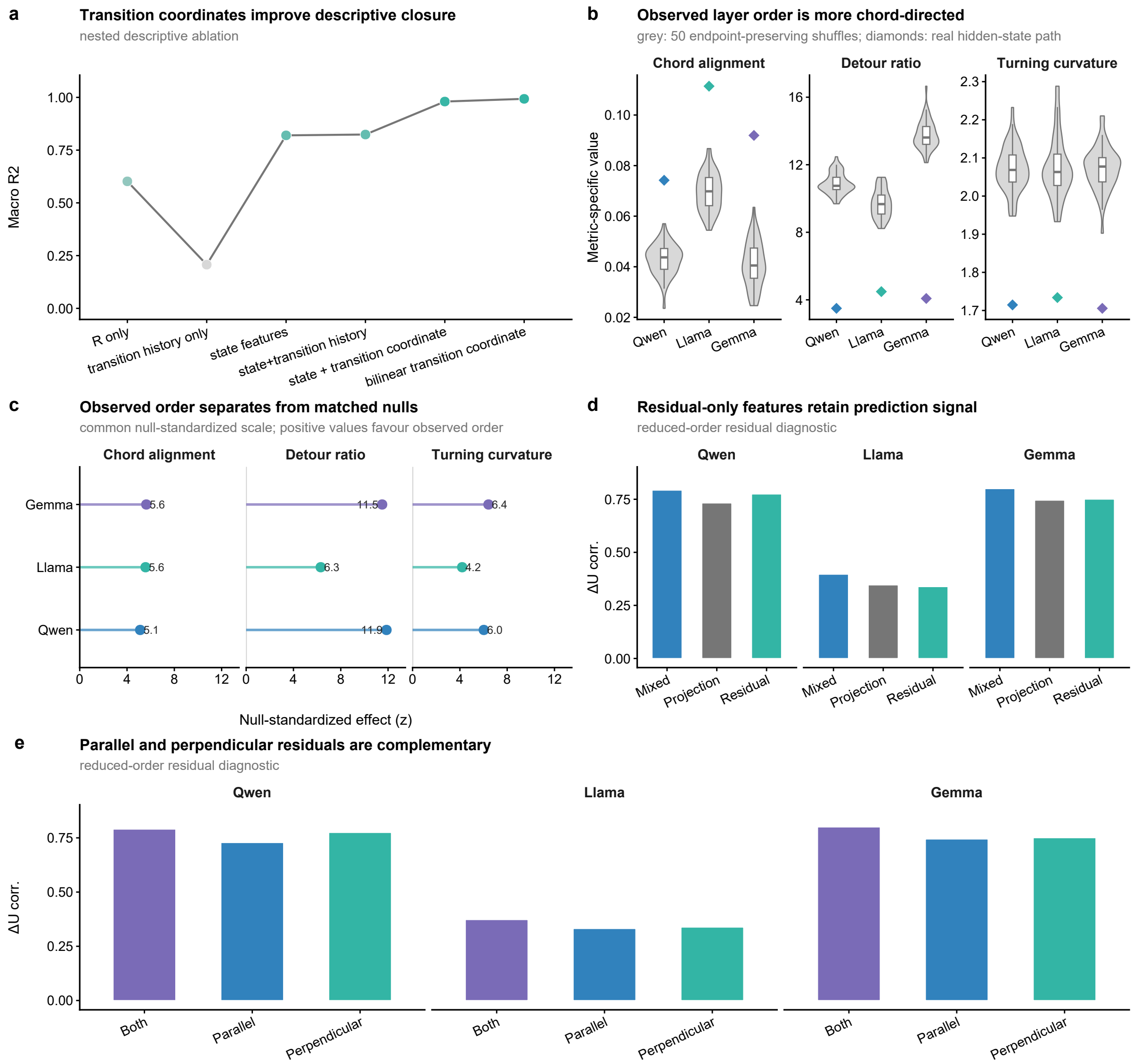


Fig. 4 | Realized transitions and layer order reveal structured flow. a, Descriptive closure for state-only, previous-transition history, state plus O_l^{cont}, and bilinear realized-transition models. When target-transition information enters O_l^{cont}, the result is descriptive rather than predictive. b, Hidden-state chord alignment for real trajectories and 50 endpoint-preserving layer-order shuffles. c, Null-standardized effects for chord alignment, detour and turning curvature on a common scale; positive values favour observed order. d, Residual-only prediction. e, Parallel and perpendicular residual proxies; these proxies are not an exhaustive tangent-normal decomposition. The geometry is metric-dependent and is not an exact-geodesic test. The transition-prediction boundary is shown in Extended Data Fig. 4. See Source Data Fig. 4 and Source Data manifest.

An actuator switch enables trajectory and output-boundary control

Observation and compact description do not by themselves establish that the trajectory account identifies an actionable process. We therefore treated intervention as a perturbational test. The first additive state-direction interventions did not produce robust mechanism-specific control. A subsequent additive flow-direction intervention also failed to produce the required selectivity (Fig. 5a). These negative results ruled out the idea that any correlated direction would serve as an actuator.

The failure prompted a change of intervention object. Adapting the control-theoretic separation between state estimation and actuator choice, we fitted a local transition operator conditioned on the predicted intervention-class label. A policy first interpolated between the realized next-layer state and the operator proposal, then applied a scaled precursor direction. The classifier distinguished predefined stable, competition and closure prompt classes; the policy selected the two weights for held-out graph groups. These classes organize the perturbation test and are not inferred dynamical phases. Final answer correctness was not a classifier target.

The resulting policy increased internal trajectory-dominance specificity over matched fixed combinations in each checkpoint (Fig. 5b). In the 50-shuffle replication, the real policy exceeded the joint mechanism-and-policy shuffle null with z scores of 4.38 for Qwen, 4.23 for Llama and 4.02 for Gemma (Fig. 5c). Because the learned actions were applied to held-out graph groups before the endpoint was measured, these contrasts provide perturbational evidence for selective causal control of the measured internal answer-formation trajectory. Mechanism macro F1 ranged from 0.827 to 0.927, whereas policy macro F1 ranged from 0.534 to 0.708, which bounds the precision of actuator assignment (Fig. 5d).

The original action budget reached the internal endpoint but not every candidate boundary. Clean-candidate selection changed by 0.0, 9.2 and 6.9 percentage points in Qwen, Llama and Gemma (Extended Data Fig. 5). Qwen nevertheless moved 86 of 87 prompts towards clean and shifted the mean margin from -5.03 to -2.54; the maximum remained -0.56. Cross-fitted ΔU-to-margin maps reached pooled $R^2 = 0.850$ and separated 15 original-budget crossings with AUROC 0.976 (Methods; Source Data). Qwen's larger deficit and 94% saturation at the maximum 0.3/0.3 action identified an exhausted budget rather than absent output leverage.

We therefore expanded the bounded action library while fitting all coordinates, transition models and boundary objectives on training graphs only. The largest operator and precursor weights were 1.0 and 1.2, versus the original 0.3/0.3 budget; this was a discrete search, not a dose-response study. Under one predeclared confidence-and-entropy gate, full-vocabulary top-1 changed from conflict to clean in 17/87, 13/87 and 9/87 held-out closure prompts in Qwen, Llama and Gemma, with 0/203, 2/203 and 0/203 non-closure changes (Fig. 5e). Among gate-open prompts that were conflict top-1 at baseline, 17/84 Qwen, 13/40 Llama and 9/17 Gemma prompts crossed. Qwen non-crossings usually moved towards clean from more distant baselines; some Llama and Gemma pairwise crossings remained behind a third token (Supplementary Methods Table 5). Fifty safety-matched reassignments preserved gate coverage and condition-by-layer action multisets; observed counts did not exceed their 95th percentiles ($P = 0.804, 0.922, 0.255$; Extended Data Fig. 6e,f). In a separately frozen component ablation, retaining the gated operator dose while setting the precursor weight to zero produced 17, 14 and 8 strict crossings; retaining only the precursor produced 0, 1 and 0. A uniform maximum action on the same gate-open sets produced 22, 15 and 10 crossings, with 0, 2 and 1 non-closure changes. Thus the local transition operator carries the principal output-boundary leverage at this operating point, whereas the precursor alone is insufficient and fine-grained dose assignment remains unsupported. Under the provenance definitions above, crossing controls the emitted token but does not imply an accuracy improvement.

We then tested transfer under a frozen mixed-arithmetic protocol. Sixty-four problems supplied six matched prompt conditions; 44 problem groups were used for task-specific fitting and 20 were held out. Model-specific intervention windows, the action library, regularization and gate thresholds were inherited without retuning. At a raw-prompt candidate-token boundary, the guarded actuator produced 2 strict crossings among 45 baseline-eligible held-out target prompts and changed 1 of 240 non-target top-1 outputs (Extended Data Fig. 9). Pooled selectivity exceeded 50 address-reassignment repeats that preserved each model's action-dose multiset (one-sided empirical $P = 0.0196$). Only Qwen, however, passed the predeclared model-level gate requiring at least five eligible prompts, a positive median candidate-margin shift and at least one strict crossing; Gemma had only four eligible prompts. This is partial actuator transfer, not confirmatory cross-task output control.

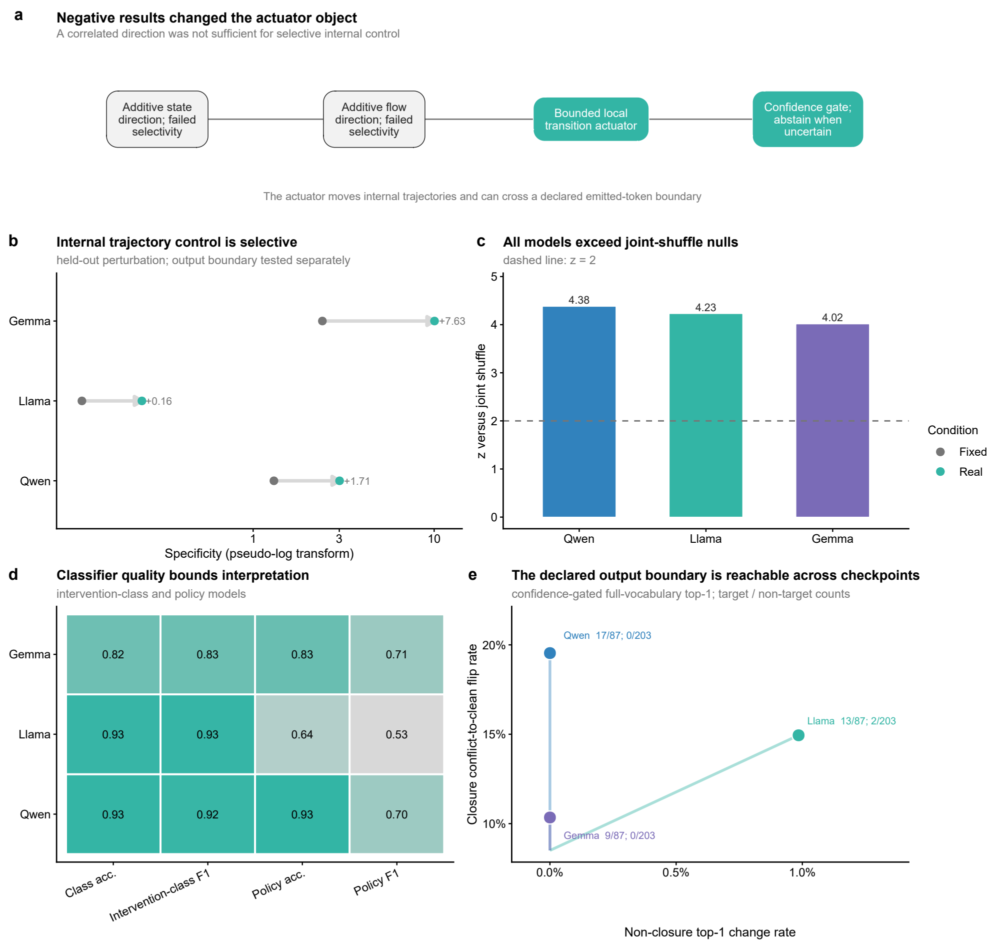


Fig. 5 | A bounded local actuator controls trajectories and a declared output boundary. a, Additive failures, bounded actuator and gate. b, Real-versus-fixed internal specificity. c, Real policy versus 50 joint-label shuffles. d, Classifier quality. e, Strict full-vocabulary conflict-to-clean crossings among 87 held-out closure prompts and collateral changes among 203 non-closure prompts: Qwen, 17/87 and 0/203; Llama, 13/87 and 2/203; Gemma, 9/87 and 0/203. Clean is provenance-defined; counts do not measure accuracy. See Extended Data Fig. 6f for action reassignment, Supplementary Methods Table 5 for non-crossings, and Source Data Fig. 5 and manifest.

Discussion

Across the tested checkpoints, realized transitions formed a measurable internal process and local perturbations crossed a declared emitted-token boundary. The new controls separate three levels that raw geometry had conflated: architecture supports relational propagation, training organizes compatible task coordinates and native layer order executes functionally consequential transitions. The ordered trajectory supplies the common empirical object on which these levels can be compared and perturbed. This is process-level causal evidence with bounded behavioural leverage, not a universal dynamical law.

The measurements draw on restricted mathematical and physical ideas. Depth indexes a discrete trajectory; checkpoint comparison resembles functional registration; ΔU adopts reaction-coordinate compression without claiming diffusion optimality, a committor or a slow variable. Transition closure tests compact representation of an observed update, whereas chord alignment, detour and turning angle test finite-path directional efficiency. The actuator separates state estimation from local action, and its gate abstains under uncertainty. These are sources for executable tests, not claims of physical identity.

The cross-disciplinary implication is a transferable audit, not a mechanism analogy. Given measurable states and transitions, ask whether a process object supports a compressive coordinate, rejects an endpoint-only explanation under an order-sensitive null, and responds at a predeclared boundary to targeted perturbation. Artificial systems make these tests unusually direct. A systems-neuroscience analogue would compare population trajectories with neuron-wise or static summaries, then perturb an inferred transition against a behavioural boundary. Neural trajectories evolve in physical time, whereas transformer depth is a computational index; appropriate states, timescales and interventions would differ. Our data neither map layers onto neural time nor imply shared mechanisms[23-25].

This view complements component-level interpretability. Circuits can identify which parts realize a local computation; probes and sparse features can identify information available at a state; readbacks can expose output-facing tendencies; and causal interventions can localize consequential variables. A trajectory-level object asks a different question: how do those local objects participate in an ordered process? The answer can then guide where component analyses and interventions should be placed.

Rival explanations remain. Proxy ranking is task- and classifier-dependent. Geometry recurred in lexical, addition and ARC tasks, but task coordinates transferred less consistently and the relation-graph ΔU axis was not reused. Prospectively tested functional windows retained class information but failed the joint three-checkpoint recurrence gate, so their locations remain exploratory. The relation-graph margin forecast did not generalize to an incremental path-memory law; target-excluded transition estimates retain only one quarter of the descriptive O_l^{cont} gain. The training audit used one small Pythia checkpoint and one lexical task, and component swaps are artificial states rather than natural training trajectories. Stronger random-state geometry, radial sensitivity and reordered executions jointly show that geometry alone is neither learned reasoning nor functional fidelity. Failure-mode decomposition showed that Qwen non-crossings were dominated by insufficient displacement from more negative starting margins, whereas Llama and Gemma also encountered third-token competition. Gated control changed 2 of 203 non-closure outputs in Llama. Component ablation located the principal leverage in the local transition operator, yet matched reassignment and uniform-action controls supported no fine-grained dose specificity. The mixed-arithmetic extension showed pooled selectivity but failed its two-model confirmation gate. These are not safety guarantees, candidate crossing is not correctness, and confirmatory output control remains bounded to the relation-graph family.

A broader LLM dynamics landscape may include architecture supplying a propagation scaffold, training organizing a task-compatible representational substrate, prompts conditioning initial states and available directions, and transformer layers realizing local transitions. The Pythia audit provides a bounded training-time bridge: fixed task coordinates emerge and function is distributed across compatible components. It does not map a complete training-shaped semantic geometry or establish how prompt-conditioned directions are formed. Those questions require new experiments rather than extrapolation from the present trajectory data.

The transferable lesson is to choose the empirical research object before its coordinates. This may guide other distributed systems whose parts are accessible but process organization remains obscure, although no transfer is claimed here. The broader question is when a chosen object makes distributed computation observable, computable and controllable. We establish one bounded artificial case; extending it towards a general theory of cognition requires concepts and experiments beyond this paper.

Scope boundary: the training audit uses one small checkpoint series and one lexical task; it separates architectural propagation from learned coordinate compatibility but does not map a complete training-shaped representational geometry.

References

1. Vaswani, A. et al. Attention is all you need. *Advances in Neural Information Processing Systems* **30**, 5998-6008 (2017).
2. Brown, T. B. et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems* **33**, 1877-1901 (2020).
3. Wei, J. et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems* **35**, 24824-24837 (2022).
4. Bubeck, S. et al. Sparks of artificial general intelligence: early experiments with GPT-4. Preprint at `https://doi.org/10.48550/arXiv.2303.12712` (2023).
5. Rogers, A., Kovaleva, O. & Rumshisky, A. A primer in BERTology: what we know about how BERT works. *Transactions of the Association for Computational Linguistics* **8**, 842-866 (2020).
6. Hewitt, J. & Liang, P. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP* 2733-2743 (2019).
7. Michel, P., Levy, O. & Neubig, G. Are sixteen heads really better than one? In *Advances in Neural Information Processing Systems* **32** (2019).
8. Elhage, N. et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread* `https://transformer-circuits.pub/2021/framework/index.html` (2021).
9. Olsson, C. et al. In-context learning and induction heads. *Transformer Circuits Thread* `https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html` (2022).
10. Meng, K. et al. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems* **35**, 17359-17372 (2022).
11. Belrose, N. et al. Eliciting latent predictions from transformers with the tuned lens. Preprint at `https://doi.org/10.48550/arXiv.2303.08112` (2023).
12. Bricken, T. et al. Towards monosemanticity: decomposing language models with dictionary learning. *Transformer Circuits Thread* `https://transformer-circuits.pub/2023/monosemantic-features/index.html` (2023).
13. Zhao, H. et al. Explainability for large language models: a survey. *ACM Transactions on Intelligent Systems and Technology* **15**, Article 20 (2024).
14. Rai, D. et al. A practical review of mechanistic interpretability for transformer-based language models. Preprint at `https://doi.org/10.48550/arXiv.2407.02646` (2024).
15. Raghu, M., Gilmer, J., Yosinski, J. & Sohl-Dickstein, J. SVCCA: singular vector canonical correlation analysis for deep learning dynamics and interpretability. *Advances in Neural Information Processing Systems* **30**, 6076-6085 (2017).
16. Kornblith, S., Norouzi, M., Lee, H. & Hinton, G. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning* 3519-3529 (2019).
17. Lange, R. D., Kwok, D., Matelsky, J. K., Wang, X., Rolnick, D. & Kording, K. Deep networks as paths on the manifold of neural representations. *Proceedings of Machine Learning Research* **221**, 102-133 (2023).
18. Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S. & Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems* **36**, 16318-16352 (2023).
19. Geva, M., Schuster, R., Berant, J. & Levy, O. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* 5484-5495 (2021).
20. Chuang, Y.-S. et al. DoLa: decoding by contrasting layers improves factuality in large language models. In *The Twelfth International Conference on Learning Representations* (2024).
21. Huben, R., Cunningham, H., Riggs Smith, L., Ewart, A. & Sharkey, L. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations* (2024).
22. Gao, L. et al. Scaling and evaluating sparse autoencoders. In *The Thirteenth International Conference on Learning Representations* (2025).
23. Chen, B., Huang, K., Raghupathi, S., Chandratreya, I., Du, Q. & Lipson, H. Automated discovery of fundamental variables hidden in experimental data. *Nature Computational Science* **2**, 433-442 (2022).
24. Chen, S. et al. Data-driven reaction coordinate discovery in overdamped and non-conservative systems: application to optical matter structural isomerization. *Nature Communications* **12**, 2548 (2021).
25. Churchland, M. M. et al. Neural population dynamics during reaching. *Nature* **487**, 51-56 (2012).
26. Clark, P. et al. Think you have solved question answering? Try ARC, the AI2 Reasoning Challenge. Preprint at `https://doi.org/10.48550/arXiv.1803.05457` (2018).

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